Linear Regression

Introduction

Linear Regression is a supervised machine learning algorithm
used to predict a continuous numerical value.

It models the relationship between an independent variable (X)
and a dependent variable (Y).

Mathematical Equation

$$Y = mX + c$$

Where:
- Y represents the predicted output.
- X represents the input feature.
- m represents the slope or coefficient.
- c represents the intercept.

Working

1. Collect labelled training data.
2. Identify input and output variables.
3. Fit a linear relationship to the training data.
4. Calculate predictions.
5. Evaluate the model using suitable metrics.

Loss Function

Mean Squared Error (MSE) measures the average squared difference
between actual and predicted values.

Applications

- House price prediction
- Sales forecasting
- Demand estimation
- Temperature prediction